HU CHONG XERN

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Education

National University of Singapore **Aug 2022 – May 2026**
Bachelor of Engineering in Computer Engineering (Second Class Upper Honours)
Specialisation in Internet of Things

Boston University, United States **Aug 2024 – Dec 2024**
Overseas Student Exchange Programme

Skills

Proficient Languages: Python, C/C++, Java, JavaScript, SQL
Frameworks/Tools: FastAPI, Springboot, Supabase, OpenCV, PyTorch, Docker, MQTT, Linux, Power BI
Domains: Data Engineering, Software Systems, Machine Learning, Internet of Things

Work Experience

Data Analyst with Machine Learning at Henkel Singapore **Jan 2025 – Jun 2025**

- Built and deployed a Python data pipeline packaged as an executable, enabling non-technical users across seven APAC manufacturing sites to run asset synchronisation analysis independently.
- Designed algorithms to detect synchronised asset behaviour and identify opportunity timestamps from historical energy data between 2021 to 2024, uncovering operational patterns and anomalies.
- Built Power BI dashboards and presented insights to stakeholders, contributing to site-level operational reviews for Bangpakong and Chonburi plants.

Software Quality Assurance Internship at Integro Technologies **May 2024 – Jul 2024**

- Investigated production issues by tracing SQL logic, test cases and system behaviour for banking clients using Oracle systems.
- Reproduced bugs and collaborated with developers to identify root causes during QA testing, communicating findings across teams using defect reports.

Project Experience

IoT Greenhouse Final Year Project **Sep 2025 – Apr 2026**

- Led end-to-end development of an IoT greenhouse system integrating sensors, computer vision, and backend pipelines for remote monitoring
- Automated periodic image processing and inference workflows via trained image segmentation model using GitHub Actions to support continuous greenhouse monitoring.
- Engineered data flow from image capture to storage and inference using Supabase, supporting continuous monitoring across 6 sensor nodes.
- Built a full-stack IoT monitoring and control platform using FastAPI REST APIs and React, enabling real-time sensor streaming, remote device control and scalable data storage through Supabase.

Embedded Computer Vision Motion Tracking Robotic System **Sep 2025 – Nov 2025**

- Deployed an AprilTag-based computer vision system using ESP32-CAM to facilitate real-time user tracking and distance computation for an autonomous mobile robot.
- Created a multi-task Convolutional Neural Network (CNN) for combined binary classification and 8-value corner regression, improving model accuracy by applying fine-tuning and custom data augmentation.
- Managed the end-to-end process of embedded camera integration, including low-latency frame streaming from camera module to the main server.

NLP Project Allocation Clustering **Apr 2025 – May 2025**

- Developed NLP-based clustering pipeline to automate categorisation of manufacturing project proposals, improving task allocation efficiency across large datasets.
- Processed and analysed hundreds of textual project descriptions for task allocation and prioritisation.

FinTrack Financial Management Application **Mar 2024 – Apr 2024**

- Developed Java-based financial management application supporting income, expense, and transaction tracking.
- Designed modular command-driven architecture using object-oriented principles, implementing structured transaction management with separate inflow/outflow abstractions.
- Built features for transaction editing, searching, categorisation and reporting, including an undo functionality for reversible operations.